

mCOLD Foam V.1 — User Manual

Version 0.1 (draft) · 2026-10-03 · firmware 0.7.0-dev

1. About mCOLD

mCOLD is a cold-chain box that records the temperature of its goods for the whole journey. From the start of a trip to its end it:

- measures the temperature with its probe and records it every 5 minutes
- raises an alarm when the temperature leaves the range set for that trip
- keeps the records on board, and sends them to the server over Wi-Fi when it can
- records its position (GNSS) and when the box is moved or knocked

2. Parts

Part	What it does
E-paper screen	Shows temperature and status; keeps its picture with no power
Three front lights	Left = goods, middle = trip, right = the box itself (see 6)
Side light	Charging status; lit only while plugged in
Buzzer	Sounds on a temperature alarm
NFC tap point	Tap a phone here to connect the app
USB-C port	Charging
Power switch	Cuts power to the whole box
Temperature probe	Measures the goods' temperature

3. Switching on and off

On: slide the switch. The three front lights sweep white from left to right, then blink green together: the box is ready. Three amber blinks on the right light instead mean a part inside is not working properly.

Off: slide the switch back. Everything stops; the clock keeps running on its backup cell. If you switch off during a trip, the trip carries on when the box is switched on again, and the gap is recorded.

On battery the box sleeps almost all the time to save power and wakes itself every 5 minutes to measure and record. The screen keeps showing the latest picture.

4. Battery and charging

Plug in USB-C to charge. The side light shows:

Side light	Meaning
Yellow, breathing	Charging
Green, breathing	Nearly full (80% and over)
Green, steady	Full
Yellow, slow blink	Power in, but not charging

Side light	Meaning
Red, blinking	Charging fault (one beep)
Off	Not plugged in

While plugged in, the box stays awake and keeps its Wi-Fi connection.

Battery protection

Battery	What the box does
Below 15%	Right light blinks amber; a low-battery alarm is recorded in the trip
Below about 10%	No new trip can start (the box keeps running) — charge first
Below about 5%	Screen shows BATTERY EMPTY and the box switches itself off, so the battery is not damaged or swollen

When the box has switched off for an empty battery, all its data is kept. Plug in USB-C: the box switches itself back on and any running trip carries on.

5. The screen

Where	Shows
Top bar	Box ID and the time (local)
Centre	Latest temperature (°C), large
Below it	Min / max of this trip, or "NO ACTIVE TRIP"
Bottom left	Wi-Fi, server, GNSS and trip icons
Bottom right	Charging, battery %, storage used %

A Wi-Fi, server or GNSS icon with a line through it is not connected, or has no recent data.

Red on the screen: red number = temperature out of range · red number and red frame = alarm · red -- = no reading from the probe.

Updates: the screen changes when its content changes, at most every 5 minutes, but at once for an alarm, a trip starting or stopping, or the cable going in or out. A refresh takes about 2 seconds.

6. Lights and sound

Light	Pattern	Meaning
All three front	White sweep, then green blink	Switched on, ready
All three front	One white blink	NFC tap received
Middle	Three green blinks	Trip started
Middle	Short green blink every 15 min	Trip running normally

Light	Pattern	Meaning
Middle	One green blink	Trip stopped
Left	Two red blinks + three beeps	Temperature alarm
Left	One blue blink	Alarm acknowledged
Right	Three amber blinks	A problem with the box: probe, low battery, clock without time

When someone lifts or moves the box, it shows its status lights briefly (a few times an hour at most, to save the battery).

7. Trips

Starting a trip

The acceptable temperature range (for example 2–8 °C) is set when each trip starts, so the same box can carry different goods. When the trip has started, the middle light blinks green three times and the screen shows the trip icon.

A trip will not start if another trip is running, the battery is too low, or the storage is full with nothing that may be deleted.

During a trip

- Temperature is recorded every 5 minutes, with battery, position and how often the box was moved.
- If the box is switched off or reset on the way, the trip carries on when it comes back; a new trip is never started in its place.

Alarms

Alarm	Raised when	Cleared when
Temperature high / low	Out of range for longer than the set time (normally 5 min)	Back inside the range by 0.5 °C
Probe	No reading from the probe for over 1 minute	Readings return
Low battery	Battery below 15%	Battery back to 20%

Acknowledging an alarm stops the sound and records that someone has seen it; the alarm history is kept in full.

Stopping a trip

The middle light blinks green once, and the screen shows the trip's summary for about an hour.

8. Connections

NFC: tap a phone on the NFC point. The front lights blink white and the box opens Bluetooth for the app for 60 seconds. The tap is also the permission: no pairing, no code.

Wi-Fi: the box remembers up to 5 networks and joins the strongest. Staff set them up. On battery it connects briefly every 5 minutes to send new records; if it cannot, it waits longer between tries to save the battery. Records waiting to be sent stay on the box.

Time: the box sets its clock from the internet or from satellites, and the clock keeps running while the box is off.

9. Data on the box

- About 268 days of records at one every 5 minutes.
- Records leave the box only after the server confirms it has stored them.
- When storage is full, the oldest trips the server has confirmed go first.
- Nothing is lost when power is cut or the battery runs out.

10. Troubleshooting

Problem	What to do
Screen shows BATTERY EMPTY / box will not switch on	Plug in USB-C; the box switches itself on
A trip will not start	Charge the battery; stop the previous trip first
Right light blinks amber	Check the probe is plugged in firmly; charge. If it persists, call staff
Screen shows a red --	The probe is unplugged or faulty: check it
Wi-Fi icon always crossed out	Out of range of the set networks; records are kept on the box
Side light blinks red	Unplug and plug in again. If it persists, call staff

Not yet included in this draft

- App steps: start/stop a trip, set the range, acknowledge alarms, view data (app not finished)
- Run time per charge and full-charge time (being measured)
- Specifications: size, weight, operating temperature, water resistance, approvals
- Pictures of the box and where the switch / port / NFC point are
- Four-colour screen (panel not in hand), shock alarm (threshold not set), door sensor (switched off)
- Reading data over USB drive / SD card, and firmware updates (deferred)
- Temperature accuracy (calibration pending)